

A4.4.2 Ethical aspects of computer technologies in daily life



A4.4.2 Discuss ethical aspects of the increasing integration of computer technologies into daily life

A4.4.2 addresses ethical aspects of increasing tech integration, emphasising reassessing guidelines for emerging technologies (e.g., AI, AR/VR, quantum computing) and their impact on society, individual rights, privacy, and equity.

Introduction

We've explored the ethics of the machine learning we use today, but what about tomorrow? Technology is advancing at an incredible pace, creating possibilities that were once the stuff of science fiction. This rapid progress means we are constantly faced with new and complex ethical questions about the kind of world we are building.

Why Ethical Guidelines Must Constantly Evolve

Technology develops much faster than laws, regulations, and social norms can keep up. The ethical rules we created for the internet of the 2000s are not sufficient for the AI of the 2020s. As new technologies emerge, they create entirely new situations and potential problems that no one had previously considered.

Therefore, it is crucial that as computer scientists and as a society, we **continually reassess our ethical guidelines**. We must be proactive in anticipating the potential impacts of new technologies on individual rights, privacy, and equity, and adapt our principles accordingly.

The Potential Implications of Emerging Technologies

Let's explore some of the most significant emerging technologies and the ethical dilemmas they present.

Pervasive AI

Pervasive AI refers to a future where artificial intelligence is seamlessly integrated into almost every aspect of our daily lives, from our homes and cars to our workplaces and public spaces.

Potential Positive Impacts:

- **Efficiency:** AI could manage city-wide systems like traffic flow and energy grids, reducing waste and congestion.
- **Personalisation:** Your environment could adapt to you, with smart homes adjusting lighting and temperature to your mood, and educational software creating a perfectly personalised learning path for every student.

Potential Ethical Risks:

- **Societal Impact (Job Displacement):** Widespread automation could lead to significant job losses in many sectors.
- **Privacy & Individual Rights:** Constant monitoring by AI systems could lead to a state of pervasive surveillance, eroding personal privacy.
- **Equity:** If access to beneficial AI is not distributed fairly, it could dramatically widen the gap between the rich and the poor.

Augmented Reality (AR)

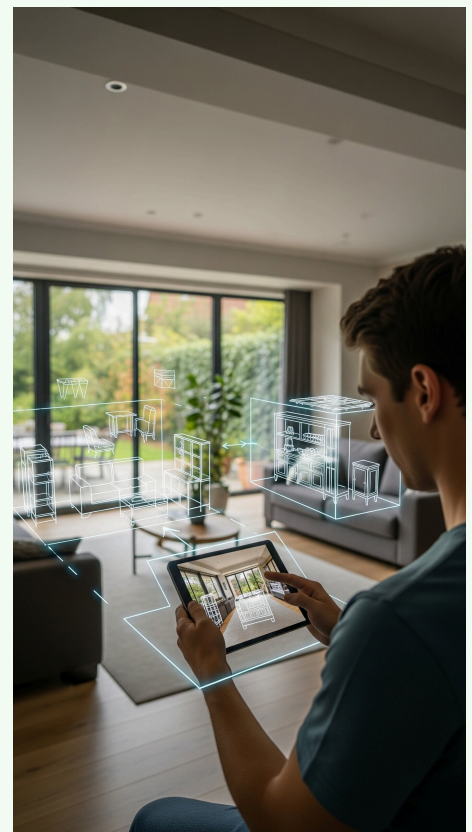
AR overlays computer-generated information onto the real world, viewed through a device like a phone or smart glasses.

Potential Positive Impacts:

- **Enhanced Information:** A surgeon could see a patient's vital signs overlaid directly on their view during an operation. An engineer could see a machine's maintenance history just by looking at it.
- **Improved Navigation & Education:** You could see directional arrows on the pavement as you walk, or see a historical reconstruction of an ancient ruin while standing in it.

Potential Ethical Risks:

- **Privacy:** If everyone is wearing AR glasses that are constantly recording, who owns that data? Can you be recorded in a public space without your consent?
- **Individual Rights & Equity:** Could AR be used to create a digital divide, where those with the technology have a significant advantage over those who don't? Could it be used to display invasive, personalized advertising everywhere you look?



Virtual Reality (VR)

VR creates a completely immersive, computer-generated simulation of an environment, shutting out the external world.

Potential Positive Impacts:

- **Immersive Training & Therapy:** Pilots can train for dangerous scenarios in a safe simulation. Therapists can use VR to help patients overcome phobias (like fear of heights) in a controlled environment.
- **Social Connection:** VR can allow people separated by vast distances to interact in a shared virtual space, feeling a sense of presence that video calls cannot replicate.

Potential Ethical Risks:

- **Societal Impact (Isolation):** If virtual worlds become more compelling than the real world, it could lead to social isolation and a breakdown of real-world communities.
- **Privacy:** A VR system can track not just what you do, but your gaze, your hesitations, and your emotional responses. This is an incredibly intimate level of data collection.
- **Equity:** Access to high-quality VR is expensive, potentially creating a new form of inequality between those who can afford immersive digital experiences and those who cannot.



Quantum Computing

Quantum computers use the principles of quantum mechanics to solve certain types of problems millions of times faster than classical computers.

Potential Positive Impacts:

- **Scientific Discovery:** They could simulate complex molecules, leading to the discovery of new medicines and materials.
- **Advanced AI:** They could train machine learning models that are far more powerful than what is possible today.

Potential Ethical Risks:

- **Security (The "Cryptopocalypse"):** The single greatest risk is that a powerful quantum computer will be able to break most of the encryption that currently protects all of our secure data—from bank accounts and government secrets to private messages. This would be a catastrophic security failure.
- **Equity:** The immense power of quantum computing will likely be controlled by a very small number of governments and corporations, giving them an unprecedented advantage and

potentially creating a massive power imbalance in the world.

Flashcards

Use this Quizlet to learn all the key terms in this section.

Flashcards

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A 4.4.2 Discuss ethical aspects of the i...

Get a hint

Pervasive AI

Click the card to flip

[View this flashcard set](#)

Choose a Study Mode

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Why is it necessary to continually reassess ethical guidelines for technology?

- A. ☐ Because old technologies stop working.
- B. ☒ Because technology advances faster than laws and social norms, creating new, unforeseen ethical dilemmas.
- C. ☐ To make the rules more complicated.
- D. ☐ To give computer scientists more freedom.

2

A city where AI manages traffic, energy, and public services to improve efficiency is an example of:

- A. ☐ Quantum Computing
- B. ☐ Augmented Reality
- C. ☒ Pervasive AI
- D. ☐ Virtual Reality

3

The single greatest security risk posed by the development of quantum computing is its ability to:

- A. ☒ Break most modern forms of encryption.
- B. ☐ Use too much electricity.
- C. ☐ Predict the stock market with perfect accuracy.
- D. ☐ Create more realistic video games.

4

A surgeon seeing a 3D model of a patient's heart overlaid on their body during a live operation is a key application of which technology?

- A. ☐ Pervasive AI
- B. ☐ Virtual Reality
- C. ☒ Augmented Reality
- D. ☐ Quantum Computing

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An ethical concern that is particularly unique to Virtual Reality (VR) is:

- A. ☐ High electricity consumption.
- B. ☐ The spread of misinformation.
- C. ☒ The collection of intimate behavioural data like gaze and emotional responses.
- D. ☐ Job displacement.

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The risk that widespread automation by AI could lead to significant job losses is primarily an issue of:

- A. ☐ Privacy
- B. ☒ Societal Impact
- C. ☐ Security
- D. ☐ Transparency

7

A major privacy concern with widespread use of AR glasses is:

- A. ☐ They might cause eye strain.
- B. ☐ They might be too expensive for most people.
- C. ☒ The potential for constant, non-consensual recording of people in public spaces.
- D. ☐ They could be used to cheat on tests.

8

Using VR to help a patient overcome a fear of flying by placing them in a simulated airplane is an example of its use in:

- A. ☐ Data analysis
- B. ☐ Gaming
- C. ☒ Therapy
- D. ☐ Social networking

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The concept of "equity" in the context of emerging technologies primarily refers to:

- A. ☐ Ensuring that all data is encrypted.
- B. ☐ Ensuring that all technologies are open source.
- C. ☒ Ensuring that the benefits and access to new technologies are distributed fairly across
- D. ☐ Ensuring that AI models are not biased.



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The rapid evolution of AI capabilities is a clear example of why:

- A. ☐ Older hardware is no longer useful.

B. ☐ We should stop developing new technologies.

C. ☐ All AI should be controlled by governments.

D. ☐ Our ethical guidelines must be continually reassessed.

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